



Federal Bridge Certification Authority (FBCA) X.509 Certificate and CRL Extensions Profile

Federal PKI Policy Authority

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Revision History

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April 21, 2026	DRAFT	Create certificate profiles for CITE testing of ML-DSA & ML-KEM algorithms starting from Draft of Common Policy profiles and C2.0 of the FBCA Profiles

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1. Introduction

This document specifies profiles **for establishing a parallel Post Quantum Cryptography (PQC) PKI in the Community Interoperability Test Environment (CITE)**. These profiles are for CITE PQC Test certificates and CRLs, using the ML-DSA and ML-KEM quantum resistant algorithms, issued under CAs cross-certified with the Federal Bridge Certification Authority (FBCA), and that have a trust path to the PQC Test Federal Common Policy CA operated by the Federal PKI Management Authority. Federal Entities must adhere to these profiles, commercial Entities must be interoperable. Certificate profiles for digital certificates whose policy OIDs cross-certify to a FBCA PIV-I policy OID must adhere to the PIV-I profiles in this document without exception.

Use of these profiles and test certificates will initially be for the FPKI community to assist in determining appropriate PQC keys and algorithms and moving forward will also be for use in updating relevant FPKI policies and documents.

Requirements are included in five sections of this document:

- Section 2: X.509 v3 Certificates
- Section 3: X.509 v2 Certificate Revocation Lists
- Section 4: Encoding of Relative Distinguished Names
- Section 5: Subject Public Key Information (SPKI)
- Section 6: Use of URIs
- Section 7: Profile Worksheets

The purpose of these profiles is to maintain consistency and interoperability across the Federal PKI trust community.

2. X.509 v3 Certificates

X.509 v3 certificates contain the identity and attribute data of the certificate subject in the base certificate fields and certificate extensions. Detailed information about X.509 certificates can be found in [X.509] and [RFC 5280].

Self-signed CA certificates must contain a non-null distinguished name in the Issuer DN that identifies the CA in a meaningful way. The Subject DN must be encoded exactly as it is encoded in the Issuer DN.

For all certificates, the base certificate fields identify the issuer (i.e., Subject DN of the Issuing CA), subject, version number, subject's public key, validity period, and certificate serial number along with the public key algorithm used to digitally sign the certificate. Certificate extensions contain additional information about the subject or the CA.

Each of the certificate profile worksheets in Section 7 list mandatory contents of a particular class of certificates. Optional features that are supported in Federal PKI are also identified. These features may be included at the discretion of the certificate issuer.

Standard certificate extensions are defined in [X.509]. For each profile worksheet, any standard extensions that are not specified as mandatory or optional must not be included.

Certificate issuers may include additional information in non-critical private certificate

extensions for local use, but should not expect clients in the Federal PKI to process this additional information. Critical certificate extensions that are not listed in these profile worksheets must not be included.

3. X.509 v2 Certificate Revocation Lists

X.509 v2 certificate revocation lists identify the issuer CA, the date the CRL was generated, the date by which the next CRL must be generated, and the list of revoked certificates.

The Certificate Revocation List worksheet in Section 6 lists mandatory contents of CRLs. Optional features that are supported in the Federal PKI are also identified. These features may be included at the discretion of the certificate issuer.

Standard CRL extensions are defined in [X.509]. For the CRL worksheet, any standard extensions that are not specified as mandatory or optional must not be included.

Certificate issuers may include additional information in non-critical private CRL extensions for local use, but should not expect clients in the Federal PKI to process this additional information. Critical CRL extensions that are not listed in the CRL worksheet must not be included in the CRLs issued.

CRLs must be stored as HTTP accessible files and may be stored as attributes in a directory.

CRLs must comply with the requirements of Section 4.9.7 of [FBCA CP] and must be full and complete as described in [RFC 5280], these CRLs must not be indirect CRLs, delta-CRLs, or CRLs partitioned by reason code.

CAs may optionally issue additional CRLs, such as CRLs partitioned by a value other than reason code or delta-CRLs.

If delta-CRLs are issued, then either the certificates or the full CRLs that correspond to the delta-CRLs should include a FreshestCRL extension that points to the delta-CRLs.

4. Encoding of Relative Distinguished Names

Certificates must use either PrintableString or UTF8String for all DirectoryString Relative Distinguished Names.

The issuer field of certificates and CRLs should be encoded exactly as it is encoded in the subject name of the signing CA certificate to avoid complications associated with name chaining and name constraints computation. Commonly used certificate path validation implementations may be unable to perform name comparisons when names are encoded using different character sets. CAs are strongly encouraged to use consistent encoding of identical distinguished name components within a hierarchy.

CAs should use consistent encoding of name constraints and all constrained name components within the certification path. Name constraints specified in CA certificates must be compared with the subject names in subsequent certificates in a certification path, to ensure they are applied correctly.

5. Subject Public Key Information (SPKI)

Algorithms Supported

These CITE Test Profiles use ML-DSA [FIPS204] for digital signature and authentication and ML-KEM [FIPS203] for key encapsulation.

Only the “pure” ML-DSA scheme is used. “Pre-hash” ML-DSA is not currently specified for use. Where pre-hashing functionality is required, the “external mu” approach described in the following clarification to FIPS 204 should be used:

<https://csrc.nist.gov/csrc/media/Projects/post-quantum-cryptography/documents/faq/fips204-sec6-03192025.pdf>

5.1. Authentication and/or Signature Certificate

All signature and/or authentication certificates shall contain one of the following OIDs in the SPKI:

id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19}
id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18}
id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17}

Parameters field shall be absent. The subject public key shall contain the byte values listed in Section 7.2, Algorithm 22 [FIPS204]; the byte values shall be raw (i.e., not ASN.1 encoded) prior to encoding as a BIT STRING.

5.2. Key Encapsulation Certificate

All key encapsulation certificates, commonly referred to as encryption certificates, shall contain one of the following OIDs in the SPKI:

id-alg-ml-kem-1024 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.3}
id-alg-ml-kem-768 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.2}
id-alg-ml-kem-512 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.1}

Parameters field shall be absent. The subject public key shall contain the byte values listed in Section 5.1, Algorithm 13, Step 19 [FIPS203]; the byte values shall be raw (i.e., not ASN.1 encoded) prior to encoding as a BIT STRING.

5.3. Signature Algorithm OIDs

A certificate or CRL shall contain one of the following values for the signature algorithm OID:

id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19}
id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18}
id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17}

Parameters field shall be absent. Actual signature shall be encoded as BIT STRING and shall contain the byte values listed in Section 7.2, Algorithm 26 in [FIPS204]. The “context string” value used when generating or verifying an ml-dsa signature on certificates and CRLs is the empty string.

6. Use of URIs

Uniform Resource Identifiers (URIs) are found in three different extensions within the certificate profiles:

- cRLDistributionPoints
- authorityInfoAccess
- subjectInfoAccess

Each of these extensions must include an HTTP URI. If an LDAP URI is included, it must appear after the HTTP URI.

For all URIs:

- The scheme portion of all URIs must be either "http" or "ldap".
- The hostname must be specified as a fully qualified domain name.
- The default port for the relevant protocol (80 for HTTP and 389 for LDAP) must be used, but need not be included in the URI.

6.1. CRL Distribution Points Extension

This section includes requirements in addition to those specified in Section 2.2.1 in [FBCA CP].

At least one HTTP URI is required and:

- Must return a file that contains the latest DER encoded full and complete CRL, with a file extension of ".crl".
- Must include “Content-Type: application/pkix-crl” in the HTTP response headers.

If the DistributionPointName is present in the issuingDistributionPoint extension of the CRL, the value must match at least one DistributionPointName in the cRLDistributionPoints extensions in each of the certificates covered by the CRL.

An LDAP URI may be included in the cRLDistributionPoints extension. If present, the LDAP URI must include the DN of the entry containing the CRL and specify the directory attribute in which the CRL is located (certificateRevocationList, authorityRevocationList, or deltaRevocationList).

6.2. Authority Information Access Extension

The HTTP URI in the authorityInfoAccess extension must contain at least one instance of

the id-ad-caIssuers access method containing a publicly accessible HTTP URI which returns a certs-only Cryptographic Message [RFC 8551] (preferred) or a single DER encoded certificate [RFC 2585] (discouraged)*. This message:

- Must not contain any self-signed CA certificates.
- Must include one or more currently valid CA certificates issued to the issuer of the certificate, which may be used to verify the signature on the certificate.
- Must be an empty certs-only CMS format, if no currently valid CA certificates can be included.

In addition, the certs-only Cryptographic Message:

- Must contain a binary file with an extension of ".p7c".
- Must include "Content-Type: application/pkcs7-mime" in the HTTP response headers.

If used, the single DER encoded certificate:

- Must have an extension of ".cer".
- Must include "Content-Type: application/pkix-cert" in the HTTP response headers.

*The use of the single DER encoded certificate option is discouraged because it does not permit zero or multiple CA certificates, thereby reducing flexibility.

An LDAP URI may be included in the authorityInfoAccess extension, id-ad-caIssuers access method, that specifies either or both the cACertificate and crossCertificatePair attributes. A CA may, alternatively, specify each of the attributes in a separate LDAP URI.

The authoritative OCSP [RFC 6960] service must be specified in the authorityInfoAccess extension, id-ad-ocsp access method, of each Subscriber certificate and the scheme portion of the URI must be "http". This HTTP response must include "Content-Type: application/ocsp-response" in the HTTP response headers.

6.3. Subject Information Access Extension

The subjectInfoAccess extension must appear in CA certificates, unless the CA certificate asserts a path length constraint of zero in the Basic Constraints extension.

When present, the subjectInfoAccess extension must contain at least one instance of the id-ad-caRepository access method containing a publicly accessible HTTP URI which returns a certs-only Cryptographic Message [RFC 8551]. This message:

- Must contain a binary (DER encoded) file with an extension of ".p7c"
- Must include "Content-Type: application/pkcs7-mime" must be included in the HTTP response headers.
- Must contain all currently valid CA certificates issued by the subject of this certificate, except self-signed certificates
- Must be an empty certs-only CMS format, if no currently valid CA certificates can be included.

An LDAP URI may be included in the subjectInfoAccess extension, id-ad-caRepository

access method. If present, the LDAP URI must include the DN of the entry containing the relevant certificates and specify the directory attribute in which the certificates are located.

6.4. Extended Key Usage (EKU)

anyExtendedKeyUsage is prohibited in all certificates.

Applicable EKUs may be added to certificate profiles based on the usage of the private key associated with the public key certificate. However, the EKUs should be added with caution since they imply privileges and capabilities to relying party systems. The following is a list of EKU OIDs that shall not be included in end entity certificates issued to humans, roles, or groups:

- OCSP Signing,
- Time Stamp,
- PIV card auth,
- PIV content signing,
- PIVI Content Signing,
- id-pkinit-KPKdc.

The above EKU OIDs can be included in NPE certificates to specifically convey the indicated capability

7. Profile Worksheets

The profile worksheets identify the mandatory and optional extensions of certificates and CRLs. Unless otherwise stated, all fields and extensions listed are mandatory. Certificate extensions defined in [RFC 5280] that are not specified as mandatory or optional in the profile worksheets must not be included.

#	Profile	Description
1	Self-Signed CA Certificate	Self-signed certificate issued by CAs primarily for establishing a trust anchor.
2	Self-Issued CA Certificate	Key rollover certificate, sometimes called a link certificate, that is self-issued by a CA but not self-signed.
3	Cross Certificate	Issued by a CA in one PKI domain to a CA in another PKI domain to enable interoperability through certificate policy mapping.
4	Intermediate/Signing CA Certificate	CA certificate issued to a subordinate CA
5	Signature Certificate	Subscriber certificate used to verify signatures.

#	Profile	Description
6	Key Encapsulation Certificate	Subscriber certificate used to perform key encapsulation for protecting symmetric keys used for encryption.
7	Authentication Certificate (non-PIV-I)	Subscriber certificate, not associated with a PIV-I credential, used to authenticate identity.
8	Device Authentication or Signature Certificate	Certificate issued to a computing or communications device (e.g., router, firewall, server) or software application for signature or authentication.
9	PIV-I Authentication Certificate	Subscriber certificate, on a PIV-I credential, used to authenticate identity. This certificate type corresponds to the PIV-I implementation of the PIV Authentication Key defined in Section 4.3 of [FIPS 201-3].
10	PIV-I Card Authentication Certificate	Subscriber certificate on a PIV-I credential, used to authenticate the PIV-I card. This certificate type corresponds to the PIV-I implementation of the Card Authentication Key defined in Section 4.3 of [FIPS 201-3].
11	PIV-I Content Signing Certificate	Certificate issued to a Card Management System for use in signing data objects on the PIV-I card.
12	Certificate Revocation List	List of revoked certificate serial numbers signed by the CA.
13	Delegated OCSP Responder Certificate	Certificates issued to OCSP responders.
14	Device Key Encapsulation Certificate	Certificate issued to a computing or communications device (e.g., router, firewall, server) or software application for protecting keys used for encryption.

Worksheet 1: Self-Signed Root Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} See 5.3.
Issuer DN	Non-null Unique DN as specified in the associated CP that identifies the CA in a meaningful way. (see Section 4 for preferred encoding).
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Must be encoded exactly as it is encoded in the Issuer DN of this certificate.
Subject Public Key Information	id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} See 5.1.
Extension	Value
Key Usage	Critical = TRUE keyCertSign, cRLSign
Basic Constraints	Critical = TRUE cA:TRUE Path length constraints should not be included.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Subject Information Access (Optional for non-Federal entities)	Must include the id-ad-caRepository (1.3.6.1.5.5.7.48.5) containing an HTTP URI pointing to a file that has an extension of .p7c. The file is a certs-only Cryptographic Message Syntax file (RFC 5751) that includes valid CA certificates issued by the subject CA. See Section 5.3 If the certificate asserts a path length constraint of zero in Basic Constraints, this extension may be omitted See 6.3.

Worksheet 2: Self-Issued CA Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the Issuing CA certificate.
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Subject DN must be encoded exactly as it is encoded in the Issuer DN of certificates and CRLs issued by this CA.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.1.
Extension	Value
Key Usage	Critical = TRUE keyCertSign, cRLSign
Extended Key Usage <i>(Optional)</i>	Included if the CA is being restricted to the issuance of specific certificate types.
Basic Constraints	Critical = TRUE cA:TRUE The pathLenConstraint field should not appear in self-issued certificates.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Information Access <i>(Optional)</i>	Must include the id-ad-caRepository (1.3.6.1.5.5.7.48.5) containing an HTTP URI pointing to a file that has an extension of .p7c. The file is a certs-only Cryptographic Message Syntax file (RFC 5751) that includes valid CA certificates issued by the subject CA. see Section 5.3 If the certificate asserts a path length constraint of zero in Basic Constraints, this extension may be omitted. See 6.3.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID as specified in Section 1.2 of the Entity CP.

Worksheet 3: Cross Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name of the Subject CA as provided in the certificate request from the Subject CA. Subject DN must be encoded exactly as it is encoded in the Issuer DN of the certificates and CRLs issued by subject CA.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.1.
Extension	Value
Key Usage	Critical = TRUE keyCertSign, cRLSign
Extended Key Usage (Optional)	Not recommended. May be included if the CA is being restricted to the issuance of specific certificate types. This may be required for inclusion in public trust stores.
Basic Constraints	Critical = TRUE cA:TRUE If the subject CA issues only subscriber certificates, the path length constraint must be present and set to zero. In all other cases, the use of a path length constraint is optional.
Subject Key Identifier	Identical to value in the Authority Key Identifier extension of the certificates issued by this CA. Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Information Access	id-ad-caRepository (1.3.6.1.5.5.7.48.5) containing an HTTP URI pointing to a file that has an extension of .p7c. The file is a certs-only Cryptographic Message Syntax file (RFC 5751) that includes valid CA certificates issued by the subject CA. If the certificate asserts a path length constraint of zero in Basic Constraints, this extension may be omitted. See 6.3.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID as specified in Section 1.2 of the Issuing CA's CP
Policy Mappings	One or more mappings from Issuing CA certificate policies to Subject CA certificate policies as determined by the Issuing domain.
Policy Constraints	Critical = TRUE requireExplicitPolicy with SkipCerts = 0 must be present. inhibitPolicyMapping must be included with SkipCerts = 0 when issued to when issued to a CA that is not a Bridge CA. inhibitPolicyMapping must be included with SkipCerts = 1 (or more). When issued to a Bridge CA SkipCerts is set to the minimum value required to support the expected mappings, usually 1.
Inhibit Any Policy	Critical = TRUE SkipCerts = 0
Name Constraints <i>(Optional)</i>	Critical = TRUE Any combination of permitted and excluded subtrees may appear. The minimum field must be zero, and maximum field must not be present.

Worksheet 4: Intermediate/Signing CA Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name of the Subject CA as provided in the certificate request from the Subject CA. Subject DN must be encoded exactly as it is encoded in the Issuer DN of the certificates and CRLs issued by the Subject CA.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.1.
Extension	Value
Key Usage	Critical = TRUE keyCertSign, cRLSign
Extended Key Usage (Optional)	Not recommended. May be included if the CA is being restricted to the issuance of specific certificate types. This may be required for inclusion in public trust stores.
Basic Constraints	Critical = TRUE cA:TRUE If the subject CA issues only subscriber certificates, the path length constraint must be present and set to zero. In all other cases, the use of a path length constraint is optional.
Subject Key Identifier	Identical to value in the Authority Key Identifier extension of the certificates issued by this CA. Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Alternative Name <i>(Optional)</i>	directoryName may be included to support local requirements
Subject Information Access	Must include the id-ad-caRepository (1.3.6.1.5.5.7.48.5) containing an HTTP URI pointing to a file that has an extension of .p7c. The file is a certs-only Cryptographic Message Syntax file (RFC 5751) that includes valid CA certificates issued by the subject CA. See Section 5.3. If the certificate asserts a path length constraint of zero in Basic Constraints, this extension may be omitted.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID as specified in Section 1.2 of the Issuing CA's CP.
Policy Constraints <i>(Optional)</i>	Critical = TRUE When this extension appears, both requireExplicitPolicy and inhibitPolicyMapping must be present and assert SkipCerts = 0.
Inhibit Any Policy <i>(Optional)</i>	Critical = TRUE SkipCerts = 0
Name Constraints <i>(Optional)</i>	Critical = TRUE Any combination of permitted and excluded subtrees may appear. The minimum field must be zero, and maximum field must be absent.

Worksheet 5: Signature Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name of the owner of the subject public key in the certificate. For requirements related to specific certificate types (e.g. PIV-I, Role-based), see [FBCACP] Section 3.1.1.1.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE digitalSignature, nonRepudiation
Extended Key Usage	One or more keyPurposeIDs consistent with digital signature must be specified. Recommended: 1.3.6.1.5.5.7.3.4 id-kp-emailProtection (required for PIV-I) 1.3.6.1.4.1.311.10.3.12 MSFT Document Signing Must not include the anyExtendedKeyUsage value.
Basic Constraints (Optional)	May be critical or non-critical cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Alternative Name <i>(Optional)</i>	rfc822Name is required if id-kp-emailProtection (1.3.6.1.5.5.7.3.4) is asserted in Extended Key Usage otherName values (e.g., Microsoft UPN) may be included to support local applications.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2. <i>Both the caIssuers and OCSP access method must be included for certificates associated with a PIV-I Card.</i>
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID contained in the Certificate Policies extension of the Issuing CA.
Subject Directory Attributes <i>(Optional)</i>	This extension may be included to indicate the cardholder's country or countries of citizenship, as specified in RFC 3739. countryOfCitizenship (1.3.6.1.5.5.7.9.4) will be an ISO 3166 Country Code(s) value.

Worksheet 6: Key Encapsulation Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name of the human owner or role/group of the subject public key in the certificate For requirements related to specific certificate types (e.g. PIV-I, Role-based), see [FBCACP] Section 3.1.1.1.
Subject Public Key Information	Must be one of the following: id-alg-ml-kem-1024 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.3} id-alg-ml-kem-768 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.2} id-alg-ml-kem-512 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.1} See 5.2.
Extension	Value
Key Usage	Critical = TRUE keyEncipherment Prohibited: All others
Extended Key Usage	One or more keyPurposeIds consistent with key management purposes must be included. Recommended: 1.3.6.1.5.5.7.3.4 id-kp-emailProtection (required for PIV-I) Must not include the anyExtendedKeyUsage value.
Basic Constraints (Optional)	May be critical or non-critical cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Alternative Name <i>(Optional)</i>	rfc822Name is required if id-kp-emailProtection (1.3.6.1.5.5.7.3.4) is asserted in Extended Key Usage otherName values (e.g., Microsoft UPN) may be included to support local applications.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2. <i>Both the caIssuers and OCSP access method must be included for certificates associated with a PIV-I Card.</i>
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID contained in the Certificate Policies extension of the Issuing CA..
Subject Directory Attributes <i>(Optional)</i>	This extension may be included to indicate the cardholder's country or countries of citizenship, as specified in RFC 3739. countryOfCitizenship (1.3.6.1.5.5.7.9.4) will be an ISO 3166 Country Code(s) value.

Worksheet 7: Authentication Certificate (Non-PIV-I)

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name of the owner of the subject public key in the certificate For requirements related to specific certificate types (e.g. Role-based), see [FBCACP] Section 3.1.1.1
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE Must assert only digitalSignature
Extended Key Usage	The following keyPurposeID values must be included: 1.3.6.1.5.5.7.3.2 TLS client authentication One or more additional keyPurposeIds consistent with authentication may be specified. For example; 1.3.6.1.4.1.311.20.2.2 Microsoft Smart Card Logon 1.3.6.1.5.2.3.4 id-pkinit-KPClientAuth 1.3.6.1.5.5.7.3.21 id-kp-secureShellClient (May only be required for administrators) Must not include the anyExtendedKeyUsage value.
Basic Constraints (Optional)	May be critical or non-critical cA:FALSE Path length constraint must be absent.

Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
Subject Alternative Name (Optional)	One or more of the following are permitted: rfc822Name otherName values (e.g., Microsoft UPN) to support local applications directoryName to support local applications
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID contained in the Certificate Policies extension of the Issuing CA.
Subject Directory Attributes (Optional)	This extension may be included to indicate the cardholder's country or countries of citizenship, as specified in RFC 3739. countryOfCitizenship (1.3.6.1.5.5.7.9.4) will be an ISO 3166 Country Code(s) value.

Worksheet 8: Device Authentication or Signature Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name that contains the name of the device in the CN.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE nonRepudiation must not be asserted in a device certificate If a certificate is used for digital signature or authentication of ephemeral keys (e.g., TLS), digitalSignature must be asserted
Extended Key Usage	May be critical or non-critical One or more key purposes consistent with the keyUsage must be specified. Must not include the anyExtendedKeyUsage value.
Basic Constraints (Optional)	cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Alternative Name <i>(Optional)</i>	The following name types may be present: dNSName is an IA5String that contains the DNS name of the subject URI is an IA5String that contains the URI of the subject rfc822Name that contains the email address of the sponsor, administrator, or help desk otherName values may also be included to support local applications
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Must assert at least one certificate policy OID designated for devices contained in the Certificate Policies extension of the Issuing CA. Device certificates should not contain policy OIDs intended for human subscribers.

Worksheet 9: PIV-I Authentication Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive INTEGER.
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049 The notAfter time must be on or before the PIV-I card expiration date.
Subject DN	Must include the CN of the subscriber using one of the name forms for PIV-I Hardware specified in [FBCACP] Section 3.1.1.1 For certificates with an Affiliated Organization: cn=Subscriber's full name, ou=Affiliated Organization Name, {Base DN} For certificates with no Affiliated Organization: cn=Subscriber's full name, ou=Unaffiliated, ou=Entity CA's Name, {Base DN} For Enterprise CAs that issue PIV-I certificates only within their own organizations, the Affiliation ou may be absent provided the organization's name appears in the {Base DN}.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE Must assert only digitalSignature
Extended Key Usage	The following keyPurposeID values must be included: 1.3.6.1.5.5.7.3.2 TLS client authentication 1.3.6.1.4.1.311.20.2.2 Microsoft Smart Card Logon One or more additional keyPurposeIds consistent with authentication purposes may be specified. For example; 1.3.6.1.5.2.3.4 id-pkinit-KPClientAuth 1.3.6.1.5.5.7.3.21 id-kp-secureShellClient (May only be required for administrators) Must not include the anyExtendedKeyUsage value.

Basic Constraints <i>(Optional)</i>	cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
Subject Alternative Name	Must include UUID. UUID uniformResourceIdentifier contains the GUID data element of the CHUID of the PIV-I Card encoded as a URN as specified in Section 3 of RFC 4122. Any additional name types may be included to support local applications. A common example is the Microsoft User Principal Name (UPN) 1.3.6.1.4.1.311.20.2.3
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Contains one certificate policy used only for PIV-I authentication certificates and mapped to 2.16.840.1.101.3.2.1.3.18. May contain additional certificate policies.
Subject Directory Attributes <i>(Optional)</i>	This extension may be included to indicate the cardholder's country or countries of citizenship, as specified in RFC 3739. countryOfCitizenship (1.3.6.1.5.5.7.9.4) will be an ISO 3166 Country Code(s) value.

Worksheet 10: PIV-I Card Authentication Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Must include the serialNumber Relative Distinguished Name set to UUID, no other name forms may be included. For certificates with an Affiliated Organization: serialNumber=UUID, ou=Affiliated Organization Name, {Base DN} For certificates with no Affiliated Organization: serialNumber=UUID, ou=Unaffiliated, ou=Entity CA's Name, {Base DN} For Enterprise CAs that issue PIV-I certificates only within their own organizations, the Affiliation ou may be absent provided the organization's name appears in the {Base DN}.
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE Must assert only digitalSignature
Extended Key Usage	Critical = TRUE Must assert only id-PIV-cardAuth keyPurposeID (2.16.840.1.101.3.6.8).
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.

Subject Alternative Name	Must include UUID. No other name forms may be included. UUID: uniformResourceIdentifier contains the UUID from the GUID data element of the CHUID of the PIV Card encoded as a URI as specified in Section 3 of RFC 4122.
CRL Distribution Points	See 6.1.
Authority Information Access	See Section 6.2 Both the caIssuers and OCSP access method must be included
Certificate Policies	Critical = FALSE Contains a single certificate policy used only for card authentication certificates and mapped to 2.16.840.1.101.3.2.1.3.19
Subject Directory Attributes <i>(Optional)</i>	This extension may be included to indicate the cardholder's country or countries of citizenship, as specified in RFC 3739. countryOfCitizenship (1.3.6.1.5.5.7.9.4) will be an ISO 3166 Country Code(s) value.

Worksheet 11: PIV-I Content Signing Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Distinguished Name indicating the organization administering the PIV-I card issuance system
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} id-ml-dsa-44 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.17} See 5.1.
Extension	Value
Key Usage	Critical = TRUE Must assert only digitalSignature
Extended Key Usage	Critical = TRUE Must assert only id-fpki-pivi-content-signing keyPurposeID (2.16.840.1.101.3.8.7)
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2. Both the caIssuers and OCSP access method must be included
Certificate Policies	Critical = FALSE Contains a single certificate policy used only for content signing certificates and mapped to 2.16.840.1.101.3.2.1.3.20

Worksheet 12: Certificate Revocation List

Field	Content
Version	INTEGER Value of "1" for Version 2 CRL.
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
This Update	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Next Update	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Revoked Certificates	The following values are included: userCertificate - the serial number of the certificate being revoked. revocationDate - the date and time of revocation. reasonCode CRL entry extension of certificateHold must be included for suspended certificates* - Use of this extension is optional for reason codes other than certificateHold. - If the revocation reason is unspecified, the reasonCode CRL entry extension should be omitted. removeFromCRL must be used only in delta CRLs. invalidityDate CRL entry extension may be included if the invalidity date precedes the revocation date. *Note: certificateHold must be used for suspension of subscriber certificates only.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
CRL Number	cRLNumber is a sequentially increasing number
Issuing Distribution Point (Optional)	Critical = TRUE This extension appears only in CRLs that do not cover all unexpired certificates in which the issuer field contains the same name as the issuer field in the CRL. For example, when a CA is rekeyed and issues separate CRLs from each key. Must conform with the requirements in section 5.2.5 of RFC 5280 with the following constraints: onlySomeReasons must not appear indirectCRL must be FALSE

Worksheet 13: Delegated OCSP Responder Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	Maximum of 120 days utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished name assigned to the OCSP Responder
Subject Public Key Information	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.1.
Extension	Value
Key Usage	Critical = TRUE Must assert digitalSignature May assert nonRepudiation
Extended Key Usage	Critical = TRUE Must assert only 1.3.6.1.5.5.7.3.9 id-kp-OCSPSigning
Basic Constraints	cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
Subject Alternative Name (Optional)	The following name types may be present: dNSName is an IA5String that contains the DNS name of the subject URI is an IA5String that contains the URI of the subject rfc822Name that contains the email address of the sponsor, administrator, or help desk otherName values may also be included to support local applications

Authority Information Access <i>(Optional)</i>	The OCSP access method must not be included. See 6.2.
Certificate Policies	Critical = FALSE Must assert all policy OIDs for which the OCSP server is authoritative.
OCSP No Check	NULL

Worksheet 14: Device Key Encapsulation Certificate

Field	Content
Version	Integer Value of 2 for Version 3 certificate
Serial Number	Unique positive integer
Signature Algorithm	Must be one of the following: id-ml-dsa-87 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.19} id-ml-dsa-65 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.3.18} See 5.3.
Issuer DN	Issuer DN must be encoded exactly as it is encoded in the Subject DN of the issuing CA certificate
Validity Period	utcTime (YYMMDDHHMMSSZ) for dates up to and including 2049 generalTime (YYYYMMDDHHMMSSZ) for dates after 2049
Subject DN	Unique Distinguished Name that contains the name of the device in the CN.
Subject Public Key Information	Must be one of the following: id-alg-ml-kem-1024 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.3} id-alg-ml-kem-768 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.2} id-alg-ml-kem-512 OBJECT IDENTIFIER ::= {2.16.840.1.101.3.4.4.1} See 5.1.
Extension	Value
Key Usage	Critical = TRUE
Extended Key Usage	May be critical or non-critical One or more key purposes consistent with the keyUsage must be specified. Must not include the anyExtendedKeyUsage value.
Basic Constraints (Optional)	cA:FALSE Path length constraint must be absent.
Subject Key Identifier	Derived using a cryptographic hash of the public key.
Authority Key Identifier	Identical to Subject Key Identifier in the issuing CA certificate. authorityCertIssuer and authorityCertSerialNumber must not be populated.
Subject Alternative Name (Optional)	The following name types may be present: dNSName is an IA5String that contains the DNS name of the subject URI is an IA5String that contains the URI of the subject rfc822Name that contains the email address of the sponsor, administrator, or help desk

	iPAddress is an octet string that contains the Internet Protocol address of the subject otherName values may also be included to support local applications
CRL Distribution Points	See 6.1.
Authority Information Access	See 6.2.
Certificate Policies	Critical = FALSE Must assert at least one certificate policy OID designated for devices contained in the Certificate Policies extension of the Issuing CA. Device certificates should not contain policy OIDs intended for human subscribers.

8. Acronyms

AKID	Authority Key Identifier
CA	Certification Authority
CMS	Cryptographic Message Syntax
CRL	Certificate Revocation List
DER	Distinguished Encoding Rules
DN	Distinguished Name
FASC-N	Federal Agency Smart Credential Number
FBCA	Federal Bridge Certification Authority
FIPS	Federal Information Processing Standards
FPKI	Federal Public Key Infrastructure
HTTP	Hypertext Transfer Protocol
LDAP	Lightweight Directory Access Protocol
NACI	National Agency Check with Inquiries
NIST	National Institute of Standards and Technology
OCSP	Online Certificate Status Protocol
OID	Object Identifier
PIV	Personal Identity Verification
PKI	Public Key Infrastructure
PKIX	Public Key Infrastructure (X.509)
RFC	Request For Comments
RSA	Rivest-Shamir-Adelman
SHA	Secure Hash Algorithm
SKID	Subject Key Identifier
S/MIME	Secure/Multipurpose Internet Mail Extensions
UPN	User Principal Name
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URN	Uniform Resource Name
UUID	Universally Unique Identifier

9. References

Please see [FBCA Certificate Policy](#) Appendix D for references..

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